Week 3 Lesson 2 — Sigmoid function

31st October 2026

1 The Sigmoid Function Clearly Explained

Notebook done to better understand The Sigmoid Function Clearly Explained

The sigmoid function is defined as

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

It is denoted by the Greek letter *sigma*.

Reference: `numpy.exp`

Calculate the exponential of all elements in the input array.

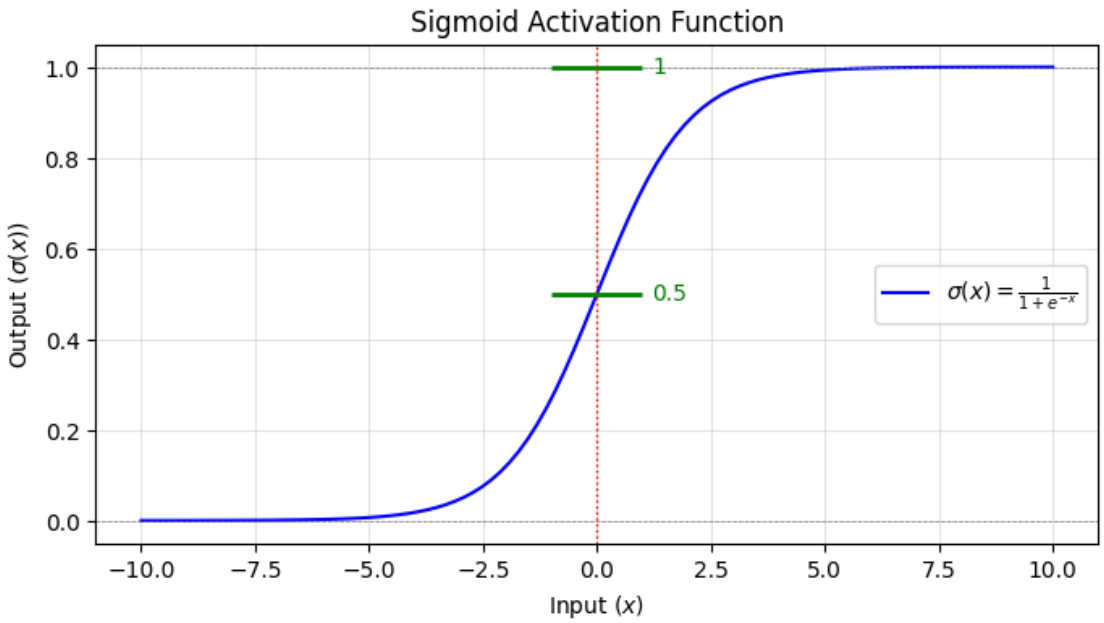
[76]:
```
import matplotlib.pyplot as plt
import numpy as np


# Define the sigmoid function
def sigmoid(x):
  return 1 / (1 + np.exp(-x))


# Generate x values from -10 to 10
x = np.linspace(-10, 10, 100)
y = sigmoid(x)

# Plot the S-shaped curve
plt.figure(figsize=(8, 4))
plt.plot(x, y, label=r"$\sigma(x) = \frac{1}{1 + e^{-x}}$", color="blue")
plt.axhline(0, color="gray", linestyle="--", linewidth=0.5)
plt.axhline(1, color="gray", linestyle="--", linewidth=0.5)
plt.axvline(0, color="red", linestyle=":", linewidth=1)
plt.title("Sigmoid Activation Function")
plt.xlabel("Input ($x$)")
plt.ylabel(r"Output ($\sigma(x)$)")
plt.grid(True)
plt.legend()
plt.hlines(y=0.5, xmin=-1, xmax=1, color="green", linewidth=2)
```

```
plt.text(1.2, 0.5, "0.5", color="green", va="center")
plt.hlines(y=1, xmin=-1, xmax=1, color="green", linewidth=2)
plt.text(1.2, 1, "1", color="green", va="center")
plt.show()
```



Exponential functions are functions of the form $f(x) = a^x$, where a is a positive constant. The sigmoid function is a special case of an exponential function.

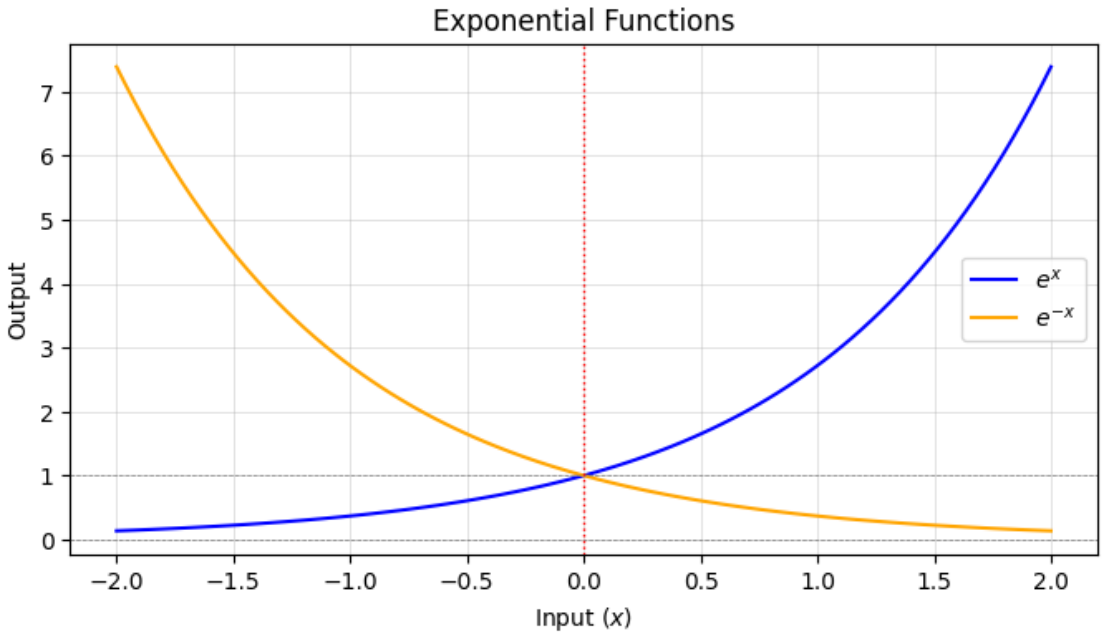
Let us plot:

$$e^{-x} = \frac{1}{e^x}$$

[77]:
```
x = np.linspace(-2, 2, 100)
y_exp = np.exp(x)
y_exp_negative = np.exp(-x)

# Plot the exponential functions
plt.figure(figsize=(8, 4))
plt.plot(x, y_exp, label=r"$e^x$", color="blue")
plt.plot(x, y_exp_negative, label=r"$e^{-x}$", color="orange")
plt.axhline(0, color="gray", linestyle="--", linewidth=0.5)
plt.axhline(1, color="gray", linestyle="--", linewidth=0.5)
plt.axvline(0, color="red", linestyle=":", linewidth=1)
plt.title("Exponential Functions")
plt.xlabel("Input ($x$)")
plt.ylabel("Output")
```

```
plt.grid(True)
plt.legend()
plt.show()
```



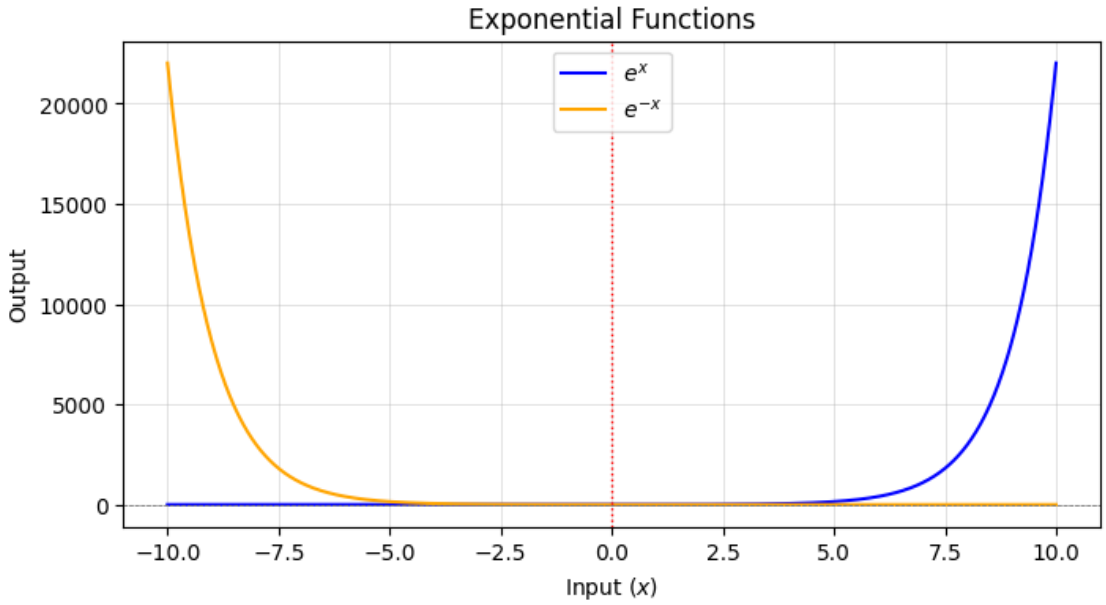
It can be seen how it grows towards infinity, and converges to 1 on x=0:

$$\lim_{x \to -\infty} e^{-x} = +\infty$$
$$\lim_{x \to 0} e^{-x} = 1$$
$$\lim_{x \to \infty} e^{-x} = 0$$

[78]:
```
x = np.linspace(-10, 10, 1001)
y_exp = np.exp(x)
y_exp_negative = np.exp(-x)

# Plot the exponential functions
plt.figure(figsize=(8, 4))
plt.plot(x, y_exp, label=r"$e^x$", color="blue")
plt.plot(x, y_exp_negative, label=r"$e^{-x}$", color="orange")
plt.axhline(0, color="gray", linestyle="--", linewidth=0.5)
plt.axhline(1, color="gray", linestyle="--", linewidth=0.5)
plt.axvline(0, color="red", linestyle=":", linewidth=1)
plt.title("Exponential Functions")
plt.xlabel("Input ($x$)")
plt.ylabel("Output")
```

```
plt.grid(True)
plt.legend()
plt.show()
```



[79]:
```
print(f"{'x':>6} | {'e^(-x)':>14}")
print("-" * 23)

for value in [-10, 0, 10]:
    print(f"{value:>6} | {np.exp(-value):>14.6g}")
```

```
     x |         e^(-x)
-----------------------
   -10 |        22026.5
     0 |              1
    10 |    4.53999e-05
```

1.1 Limits of the Sigmoid

$$\lim_{x\to-\infty}\sigma(x)=\lim_{x\to-\infty}\frac{1}{1+e^{-x}}=\frac{1}{1+\infty}=0$$
$$\lim_{x\to0}\sigma(x)=\lim_{x\to0}\frac{1}{1+e^{-x}}=\frac{1}{1+1}=0.5$$
$$\lim_{x\to\infty}\sigma(x)=\lim_{x\to\infty}\frac{1}{1+e^{-x}}=\frac{1}{1+0}=1$$

The sigmoid function has a **squashing effect**, mapping any input value to a value between 0 and 1. This makes it useful for binary classification problems, where the output can be interpreted as the probability of the input belonging to a certain class.

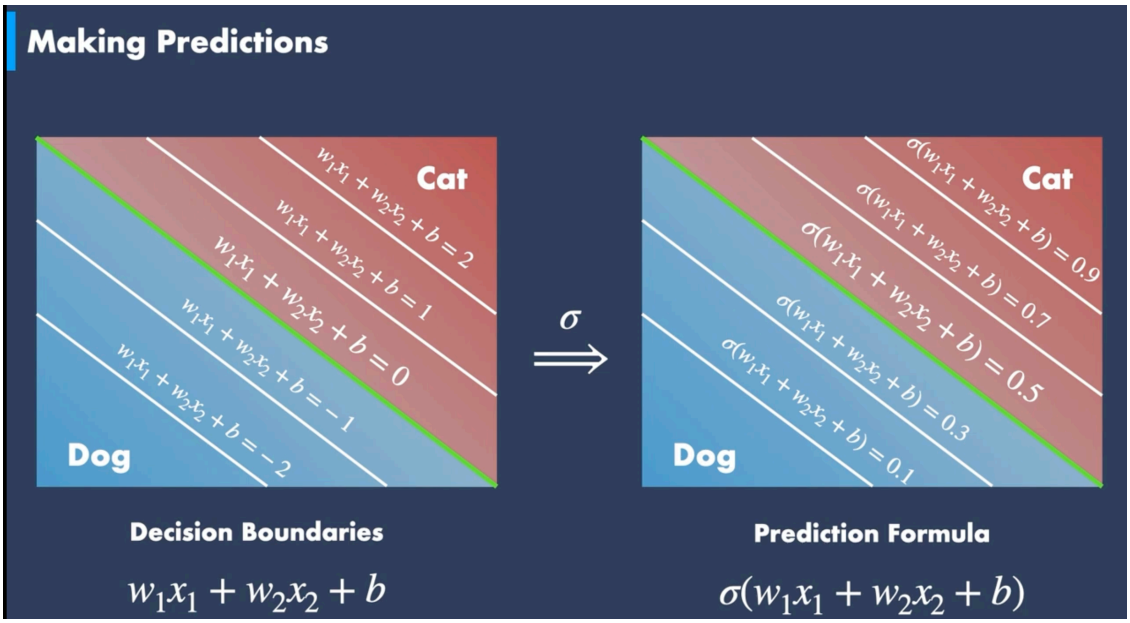
What this means is:

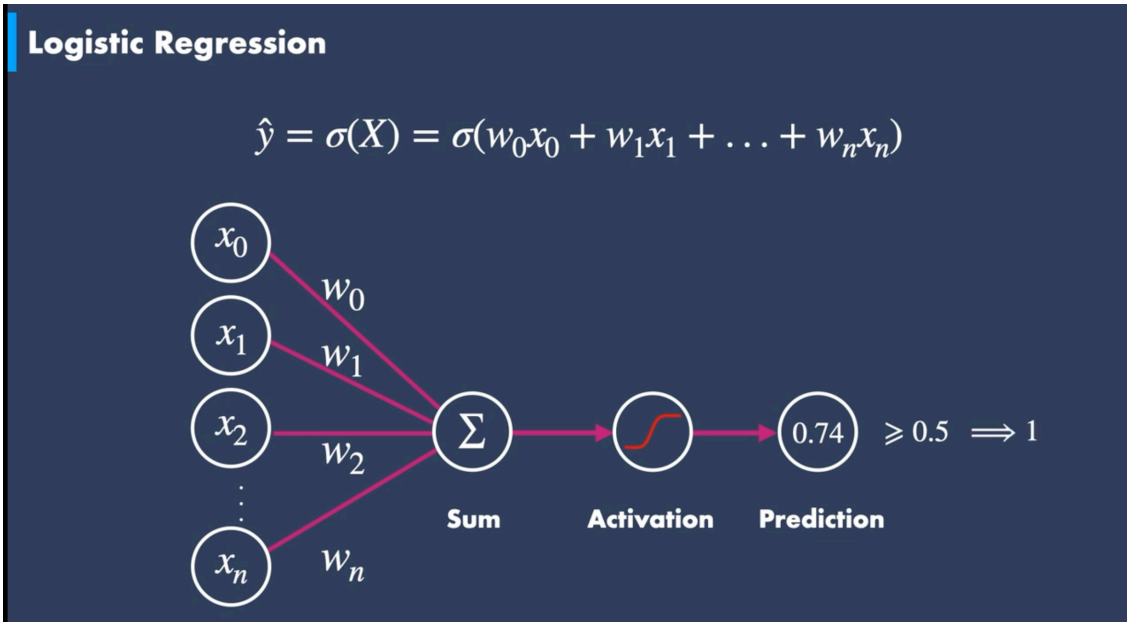
$$-\infty \leq x \leq \infty \Rightarrow 0 < \sigma(x) < 1$$

For any value that x takes, the value of the sigmoid function is always between zero and one.

$$x \in \mathbb{R} \Rightarrow 0 < \sigma(x) < 1$$

1.2 Making predictions





If it's more than 0.5 we assign it a value of 1. If it's less, a value of 0.

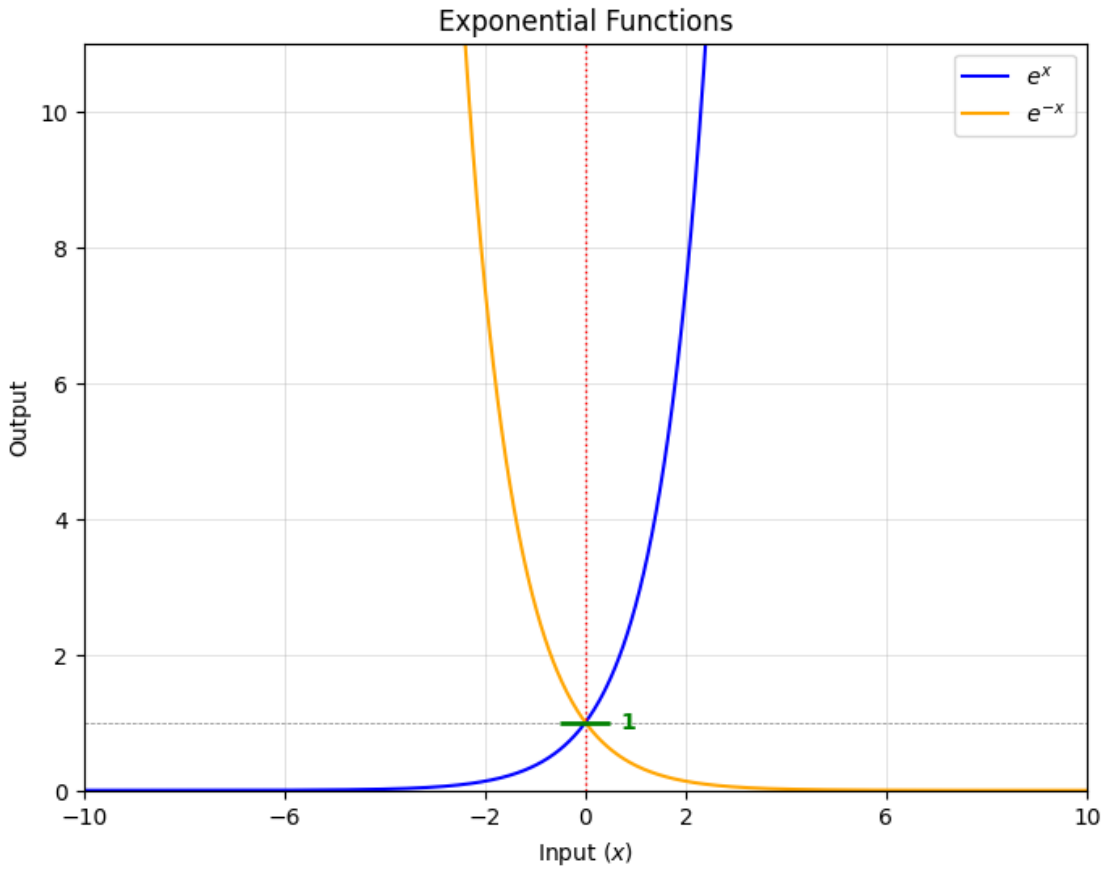
1.3 Variations of the graph

In part of the video the e^x and e^{-x} seems to be deformed to make it more visual. Let me play with the functions.

The video uses a wider x-range (−10 to 10) and a limited y-range, which makes the curves look narrower. You can replicate that without changing the functions.

[80]:
```
x = np.linspace(-10, 10, 1001)
y_exp = np.exp(x)
y_exp_negative = np.exp(-x)

# Plot the exponential functions
plt.figure(figsize=(8, 6))
plt.plot(x, y_exp, label=r"$e^x$", color="blue")
plt.plot(x, y_exp_negative, label=r"$e^{-x}$", color="orange")
plt.axhline(0, color="gray", linestyle="--", linewidth=0.5)
plt.axhline(1, color="gray", linestyle="--", linewidth=0.5)
plt.axvline(0, color="red", linestyle=":", linewidth=1)
plt.title("Exponential Functions")
plt.xlabel("Input ($x$)")
plt.ylabel("Output")
plt.grid(True)
plt.legend()
plt.xlim(-10, 10)
plt.ylim(0, 11)
plt.xticks([-10, -6, -2, 0, 2, 6, 10])
plt.hlines(y=1, xmin=-0.5, xmax=0.5, color="green", linewidth=2)
plt.text(0.7, 1, "1", color="green", weight="bold", va="center")
plt.show()
```



1.4 From the module 3

Figure recreated in my notebook.

Why this illustration uses two sets of coefficients: In the example created with the assistant, the coefficients were chosen separately to give a clear visual comparison resembling the reference figure:

- Linear model: $b_0 = -0.1$, $b_1 = 0.12$.
- Logistic model: $b_0 = -5$, $b_1 = 1$.

These illustrative values were not fitted to data. They make both curves pass through $(5, 0.5)$, while showing a gradual blue line and a steeper red S-shaped curve. Although the labels use the same generic symbols b_0 and b_1, their values differ between the two models in this plot. Therefore, the red curve is not the sigmoid of the displayed blue line.

To show the direct transformation instead, use one shared set of coefficients: calculate the linear score $z = b_0 + b_1 x$, then calculate $p = \sigma(z) = 1/(1 + e^{-z})$. The example below headed "Using the same coefficients" demonstrates this relationship.

[81]:
```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, 10, 500)

# Linear model
b0_linear = -0.1
b1_linear = 0.12
y = b0_linear + b1_linear * x

# Logistic model
b0_logistic = -5
b1_logistic = 1
p = 1 / (1 + np.exp(-(b0_logistic + b1_logistic * x)))

fig, ax = plt.subplots(figsize=(10, 5))

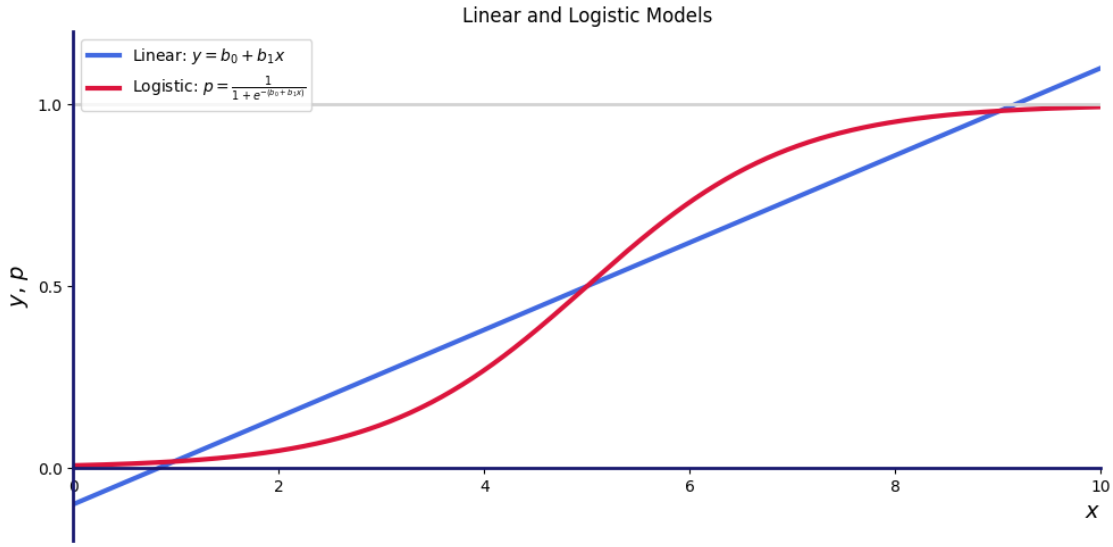
ax.plot(x, y, color="royalblue", linewidth=3,
        label=r"Linear: $y = b_0 + b_1x$")
ax.plot(x, p, color="crimson", linewidth=3,
        label=r"Logistic: $p = \frac{1}{1 + e^{-(b_0 + b_1x)}}$")

# Reference lines and axes
ax.axhline(1, color="lightgray", linewidth=2)
ax.spines["bottom"].set_position(("data", 0))
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

for side in ["left", "bottom"]:
    ax.spines[side].set_color("midnightblue")
    ax.spines[side].set_linewidth(2)

ax.set_xlim(0, 10)
ax.set_ylim(-0.2, 1.2)
ax.set_yticks([0, 0.5, 1])
ax.set_xlabel("$x$", loc="right", fontsize=14)
ax.set_ylabel("$y$, $p$", fontsize=14)
ax.set_title("Linear and Logistic Models")
ax.legend(loc="upper left")

plt.tight_layout()
plt.show()
```



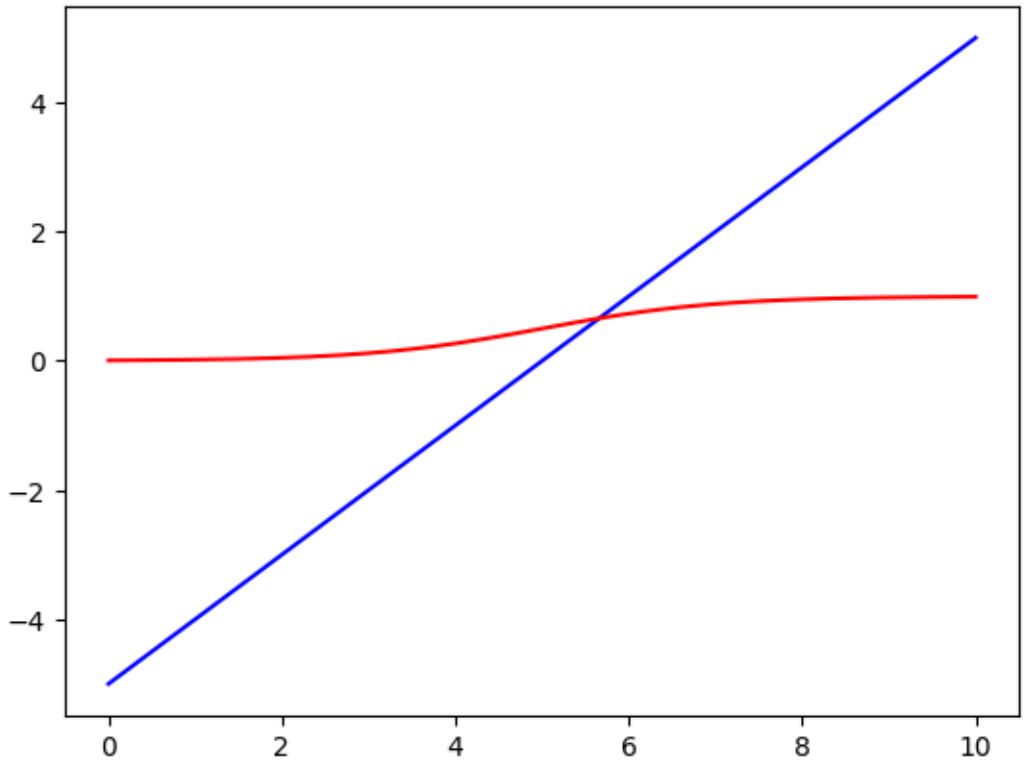
Using the same coefficients:

[82]:
```
b0 = -5
b1 = 1

linear_score = b0 + b1 * x
p = 1 / (1 + np.exp(-linear_score))

plt.plot(x, linear_score, color="blue", label="Linear score")
plt.plot(x, p, color="red", label="Logistic probability")
```

[82]:
```
[<matplotlib.lines.Line2D at 0x171c2a420>]
```


4
2
0
−2
−4
0
2
4
6
8
10